CariSurg MedTech Pathways Research Project

Project Title: Development of an Interpretable Digital Triage System for Dynamic Risk Assessment in Emergency Care

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Statement of the Problem:

Emergency department overcrowding delays care for critically ill patients because traditional triage relies on static, point-in-time assessments that fail to capture real-time physiological deterioration in the waiting room. Current AI triage solutions operate as uninterpretable "black boxes," hindering clinical adoption and alignment with established hospital protocols. This 12-week pilot introduces a transparent digital triaging model that translates real-time patient data into actionable risk categories using recognized NEWS2 and SIRS frameworks, safely accelerating clinical throughput and mitigating under-triage errors.

Previous Work:

Extensive peer-reviewed literature establishes that artificial intelligence can significantly optimise emergency department workflows. [12] conducted a systemised scoping review across EMBASE, MEDLINE, and Web of Science and found that machine learning models consistently demonstrated superior discrimination and predictive accuracy compared to traditional triage systems, successfully enhancing disease identification, predicting hospital admission needs, and improving resource allocation. Critically, the review also flagged that

AI algorithms risk producing overconfident outputs, which poses direct patient safety concerns and underscores the need for interpretable rather than purely high-performing models. [11] reinforced this multimodal direction, finding through a narrative synthesis of 20 peer-reviewed papers that combining free-text nurse notes with structured vital sign data consistently improved predictive performance over structured data alone, with NLP models achieving high accuracy in predicting triage scores, hospital admissions, and critical illness. [1] further demonstrated this through the TriageIntelli framework, confirming that fusing heterogeneous clinical data streams yields superior predictive accuracy compared to single-modality baselines, though they noted that heavy data dependency and variable performance across settings remain significant implementation barriers.

The interpretability of these systems has emerged as a distinct and critical research focus. [3] conducted a systematic PRISMA-guided review of ten empirical studies measuring the direct impact of explainable AI on clinician trust and found that XAI can significantly increase trust compared to black-box models, but only when explanations are concise, clear, and relevant to actual clinical workflows. Importantly, their analysis revealed that complex or contradictory explanations can actively undermine trust or cause clinicians to accept flawed advice, proving that the design of interpretability is as consequential as its presence. This finding directly informs the design decision in this project to map outputs onto NEWS2 and SIRS frameworks rather than presenting raw algorithmic scores.[13] demonstrated the practical viability of this approach, developing an interpretable machine learning triage tool using 323,509 ED episodes in Singapore that

achieved an AUROC of 0.823 for 30-day mortality prediction while outperforming all six established clinical benchmarks including NEWS and MEWS, confirming that interpretability and clinical-grade accuracy are not mutually exclusive goals. However, significant gaps persist in the existing literature. [10] identified the most critical structural limitation, noting that the majority of current AI triage models evaluate patients using a single snapshot taken at arrival and completely ignore physiological deterioration during prolonged waiting room stays. [8] quantified the cost of this limitation directly, demonstrating that dynamic early warning scores incorporating temporal vital sign trends outperformed static NEWS2 in predicting ICU admission or death within 24 hours (AUROC 0.906 versus 0.857) and urgent deterioration within four hours (AUROC 0.877 versus 0.829), catching more critical events with the same rate of false alarms. [6] identified the second persistent gap, finding through a narrative review of 26 peer-reviewed studies that even high-performing AI triage systems consistently fail to reach clinical deployment because uninterpretable outputs fuel clinician scepticism and prevent alignment with standardised hospital protocols. Two critical gaps therefore emerge from this body of literature and directly shape the design of this pilot. The first is the near-universal reliance on static, single-point triage assessment, with the majority of current models capturing one snapshot at arrival and failing entirely to monitor physiological trajectory during waiting room stays [8][10]. The second is the persistent failure of even high-performing AI triage systems to reach clinical deployment, driven by uninterpretable outputs that prevent alignment with standardised hospital protocols and erode clinician trust [3][6]. This pilot directly addresses both gaps.

Literature Review:

Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Lee, D. W., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2024). Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. Cureus, 16(5), e59906.

Clinical/Engineering Problem: Emergency department (ED) triage lacks precision and certainty regarding its overall impact when using traditional systems. There is a critical need to improve earlier diagnosis, patient risk prioritization, resource allocation (like predicting length of stay), and determining which patients accurately require hospitalization. ***Method***

Used: A systemized literature review was conducted in September 2023 across the EMBASE, Ovid MEDLINE, and Web of Science databases.
Reported Outcome: Machine learning models consistently showed superior discrimination and predictive accuracy compared to traditional triage systems. The algorithms successfully enhanced disease identification, accurately determined when urgent patients needed hospital admission, and improved hospital resource allocation and overall quality of care. ***Stated Limitation:*** The text notes that there is still little known overall about the use of AI in ED triage, and it highlights a specific danger where AI algorithms can give "overconfident answers" that could pose safety risks to patients.

Da'Costa, A., Teke, J., Origbo, J. E., Osonuga, A., Egbon, E., & Olawade, D. B. (2025). AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. International Journal of Medical Informatics, 197, 105838.

Clinical/Engineering Problem: Emergency Departments (EDs) face severe operational pressure from overcrowding, resource constraints, and traditional triage systems that rely on
subjective assessments, leading to a lack of consistency in patient prioritization during peak hours or mass casualty events.
Method Used: A narrative review was conducted analyzing peer-reviewed articles published between 2015 and 2024. The papers were identified through targeted searches across PubMed, Scopus, IEEE Xplore, and Google Scholar databases, with the findings synthesized into a comprehensive overview.
Reported Outcome: The synthesized findings showed substantial benefits for AI-driven triage, including improved patient prioritization, reduced wait times, and optimized hospital resource allocation, demonstrating a strong potential to transform ED operations and improve patient outcomes in high-pressure environments.
Stated Limitation: Widespread adoption is significantly blocked by operational and technical challenges, specifically data quality issues, algorithmic bias, a lack of clinician trust, and unresolved ethical concerns regarding equitable implementation.

Araouchi, Z., & Adda, M. (2024). TriageIntelli: AI-Assisted multimodal triage system for health centers. Procedia Computer Science, 251, 430–437.

Clinical/Engineering Problem: Traditional emergency department triage methods—like the Emergency Severity Index (ESI) and the Canadian Triage and Acuity Scale (CTAS)—suffer from human subjectivity, cannot handle severe overcrowding efficiently, and lead to poor hospital resource allocation.
***Method Used*:** A literature review examining and comparing the evolutionary transition of emergency triage systems from conventional, manual protocols over to modern AI-assisted predictive models.
Reported Outcome: The review found that AI models offer enhanced diagnostic precision,

optimized patient prioritization, and a reduction in human error, which collectively streamlines busy emergency room workflows.
Stated Limitation: The implementation of these models faces critical bottlenecks, specifically heavy data dependency, unresolved ethical challenges, and inconsistent, variable performance when deployed across different healthcare settings.

Sitthiprawiat, P., Wittayachamnankul, B., Sirikul, W., & Laohavisudhi, K. (2025). Development and internal validation of an AI-based emergency triage model for predicting critical outcomes in emergency department. Scientific Reports, 15(1), 31212.

Clinical/Engineering Problem: Severe emergency department (ED) overcrowding delays patient care and worsens clinical outcomes because traditional triage systems have inherent limitations in accuracy and consistency when identifying who needs urgent resources.
Method Used: A retrospective analysis of 163,452 ED visits (from 2018 to 2022) compared logistic regression, random forest, and XGBoost machine learning models against the traditional Canadian Triage and Acuity Scale (CTAS). The models evaluated mode of arrival, patient age, and free-text chief complaints using advanced multilingual sentence embeddings.
Reported Outcome: The XGBoost model achieved superior predictive performance for ICU admission and resource needs over traditional CTAS (AUROC 0.917 vs. 0.882). Incorporating unstructured text data via machine learning significantly enhanced triage accuracy and resource allocation by identifying critically ill patients more effectively. ***Stated Limitation:*** The study relied on a retrospective analysis of historical data from a single facility (Maharaj Nakhon Chiang Mai Hospital), meaning the model was validated internally and its performance may vary when deployed in a live, real-time clinical environment.

Stewart, J., Lu, J., Goudie, A., Arendts, G., Meka, S. A., Freeman, S., Walker, K., Sprivulis, P., Sanfilippo, F., Bennamoun, M., & Dwivedi, G. (2023). Applications of natural language processing at emergency department triage: A narrative review. PLoS ONE, 18(12), e0279953.

Clinical/Engineering Problem: Standard emergency triage risk stratification techniques struggle to predict clinical outcomes like hospital admission and critical illness because they completely ignore the rich, unstructured human language contained within free-text nursing notes.
Method Used: A scoping review and narrative synthesis screened 3,730 studies across MEDLINE, Embase, Cochrane, Web of Science, and Scopus to evaluate 20 peer-reviewed papers that applied NLP to free-text data acquired at ED triage. The researchers used the Prediction model Risk of Bias Assessment Tool (PROBAST) to systematically evaluate the quality and bias of the included predictive models.
Reported Outcome: NLP models achieved high accuracy in predicting triage scores, hospital admissions, and critical illness. Furthermore, combining free-text data with structured data consistently improved model predictive performance compared to using structured data

alone.

Stated Limitation: The findings are heavily restricted because 80% of the reviewed studies carry a high risk of bias, almost all the research is retrospective, and there is a critical shortage of prospective clinical trials or real-world model deployment in live practice.

Rosenbacke, R., Melhus, Å., McKee, M., & Stuckler, D. (2024). How Explainable Artificial Intelligence Can Increase or Decrease Clinicians' Trust in AI Applications in Health Care: Systematic Review. JMIR AI, 3, e53207. https://doi.org/10.2196/53207

Clinical/Engineering Problem: The "black box" nature of clinical artificial intelligence creates severe user barriers. If clinicians cannot understand an algorithm's reasoning, they may completely reject its advice. Conversely, if they trust it blindly, they may defer to incorrect machine outputs that conflict with their own medical judgment, leading to dangerous clinical errors.

Method Used: A systematic review was conducted following the PRISMA guidelines, searching the PubMed and Web of Science databases. The researchers screened 778 articles to identify 10 empirical studies that measured the direct impact of Explainable AI (XAI) on clinicians' trust using specific cognition- or affect-based trust metrics. Each included paper was systematically assessed for methodological risk of bias.

Reported Outcome: The review confirmed that XAI can significantly increase clinician trust compared to "black box" AI, but *only* if the generated explanations are concise, clear, and relevant to actual clinical workflows. It also revealed that providing explanations does not
automatically improve trust, proving that trust can be carefully modulated and balanced through smart XAI system design.

Stated Limitation: All ten evaluated empirical studies carried a moderate to high risk of bias. Furthermore, the existing literature relies heavily on abstract, cognitive definitions of trust rather than real-world behavioral measures, and the review notes that complex or contradictory explanations can actively undermine clinical trust or cause clinicians to accept flawed advice.

Xie, F., Ong, M. E. H., Liew, J. N. M. H., Tan, K. B. K., Ho, A. F. W., Nadarajan, G. D., Low, L. L., Kwan, Y. H., Goldstein, B. A., Matchar, D. B., Chakraborty, B., & Liu, N. (2021). Development and Assessment of an Interpretable Machine Learning Triage Tool for Estimating Mortality After Emergency Admissions. JAMA network open, 4(8), e2118467. https://doi.org/10.1001/jamanetworkopen.2021.18467

Clinical/Engineering Problem: Emergency department triage requires highly complex clinical judgment. While existing scoring tools are meant to assist with risk stratification, they have significant built-in limitations and struggle to provide early, highly accurate estimates of a patient's risk of mortality using a small, practical set of variables.

Method Used: A single-site, retrospective cohort study of 323,509 total ED episodes (split into training, validation, and testing cohorts) for patients subsequently admitted to a tertiary hospital in Singapore. The researchers used an interpretable machine learning framework to develop the Score for Emergency Risk Prediction (SERP) tool based on a stripped-down (parsimonious) list of triage variables, comparing its predictive accuracy for 2-, 7-, and 30-day mortality against six established clinical benchmarks (including

NEWS and MEWS).

Reported Outcome: The machine learning-derived SERP tool significantly outperformed all traditional clinical benchmark scores across all target timelines, achieving strong predictive power (such as an AUROC of 0.823 for 30-day mortality) while maintaining easy implementation and simple data collection in a fast-paced ED.

Stated Limitation: The tool was derived and tested using a retrospective cohort study from a single hospital site, meaning its real-world clinical performance still needs to be widely applied and validated across different types of healthcare settings and live clinical environments.

Gonem, S., Taylor, A., Figueredo, G., Forster, S., Quinlan, P., Garibaldi, J. M., McKeever, T. M., & Shaw, D. (2022). Dynamic early warning scores for predicting clinical deterioration in patients with respiratory disease. Respiratory research, 23(1), 203. https://doi.org/10.1186/s12931-022-02130-6

Clinical/Engineering Problem: The standard National Early Warning Score 2 (NEWS-2) framework fails to account for fine-grained clinical details or how a patient's vital signs change over time (temporal trends), which limits its accuracy when trying to predict sudden health decline or urgent treatment needs.

Method Used: Data-driven time-series features from electronic records across 31,590 respiratory in-patient episodes (from 2015 to 2020 at a large UK hospital trust) were extracted. Researchers manually annotated 1,100 of these episodes to pinpoint the exact timing of clinical deterioration and fed these trend variables into logistic regression models to build a Dynamic Early Warning Score (DEWS), testing its predictive accuracy against standard NEWS-2.

Reported Outcome: The dynamic model (DEWS) significantly outperformed standard NEWS-2 in predicting intensive care unit admission or death within 24 hours (AUROC 0.906 vs. 0.857) and urgent clinical deterioration within 4 hours (AUROC 0.877 vs. 0.829). DEWS caught more critical events ahead of time while maintaining the same low rate of false alarms.

Stated Limitation: The study relied entirely on historical data from respiratory patients at a single acute NHS Trust. The authors explicitly stated that prospective validation studies are still required to see if these dynamic scores actually reduce missed deteriorations and false alarms in live, real-world clinical environments.

Boulitsakis Logothetis, S., Green, D., Holland, M., & Al Moubayed, N. (2023). Predicting acute clinical deterioration with interpretable machine learning to support emergency care decision making. Scientific reports, 13(1), 13563. https://doi.org/10.1038/s41598-023-40661-0

Clinical/Engineering Problem: Increasing operational pressures in fast-paced Emergency Departments create an urgent need to efficiently identify patients at risk of rapid health decline. Traditional tracking frameworks like NEWS2 can cause alert fatigue or entirely miss high-risk patients who present with low initial scores.

Method Used: A retrospective study analyzing a dataset of 118,886 unplanned hospital admissions at Salford Royal Hospital, UK. The researchers systematically compared machine

learning algorithms (logistic regression, gradient-boosted decision trees, and support vector machines) using a patient's first recorded vital signs, lab results, and observations to predict mortality or critical care needs within 24 hours. The team used Shapley Additive exPlanations (SHAP) to map out feature weights and justify model outputs. ***Reported Outcome:*** The machine learning models significantly outperformed the traditional NEWS2 framework, yielding up to a 0.366 increase in average precision, a notable reduction in daily false alarms (alert fatigue), and a drop in biased predictions across different age and sex demographics.
Stated Limitation: The models were built on historical, cross-sectional data from a single hospital site, and the authors explicitly noted that further work is still needed to trial and test these models in actual, live clinical practice.
What it Proves: This study directly evaluates machine learning algorithms (such as gradient-boosted decision trees) against the traditional NEWS2 framework using Electronic Health Record (EHR) data from ED patients.

Liu, N., Xie, F., Siddiqui, F. J., Ho, A. F. W., Chakraborty, B., Nadarajan, G. D., Tan, K. B. K., & Ong, M. E. H. (2022). Leveraging Large-Scale Electronic Health Records and Interpretable Machine Learning for Clinical Decision Making at the Emergency Department: Protocol for System Development and Validation. JMIR research protocols, 11(3), e34201. https://doi.org/10.2196/34201

Clinical/Engineering Problem: Growing global demand and overcrowding in Emergency Departments lead to longer waiting times. While machine learning tools can build triage and risk prediction models to help, the "black box" nature of these advanced models limits their actual clinical use and interpretation by staff.
Method Used: A retrospective, single-center study protocol analyzing a massive longitudinal dataset of approximately 1.8 million ED visits (representing over 810,000 unique patients) from a tertiary hospital in Singapore. The researchers used a machine learning tool called *AutoScore* to create three dynamic, interpretable scores (called SERT) to evaluate patients at different times: on arrival, during their ED stay, and at admission.
Reported Outcome: The system successfully creates a unique scoring system that is both dynamic and transparent. It establishes a clear standard for using large-scale Electronic Health Records (EHRs) and interpretable machine learning to predict severe outcomes, like the need for an intensive care unit (ICU) or inpatient death.
Stated Limitation: The paper describes a retrospective study protocol and system design rather than a finished prospective clinical trial, meaning the models still require further real-world validation to prove their clinical safety and impact in live practice.
What it Proves: The authors outline the clinical imperative for moving away from 5-level static tools (like the Emergency Severity Index) and standard scores because they are subjective, static, and fail to track patients whose conditions rapidly change or deteriorate during an ED stay.

Chang, Y. H., Lin, Y. C., Huang, F. W., Chen, D. M., Chung, Y. T., Chen, W. K., & Wang, C. C. N. (2024). Using machine learning and natural language processing in triage for prediction of clinical disposition in the emergency department. BMC emergency medicine, 24(1), 237. https://doi.org/10.1186/s12873-024-01152-1

Clinical/Engineering Problem: Accurate triage is essential to allocate hospital resources

efficiently and reduce overall patient stay times. However, manual triage decisions are frequently subjective and vary significantly from one medical provider to another, leading to dangerous errors where patients are either over-triaged or under-triaged.
Method Used: A multi-center retrospective study using a large-scale dataset of 213,984 total non-trauma adult ED visits across two separate institutions (China Medical University Hospital for internal training and Asia University Hospital for external validation). The researchers combined structured data with unstructured free-text nursing notes processed via Natural Language Processing (NLP) to train six distinct machine learning algorithms. They directly compared the models' ability to predict patient hospital disposition using actual emergency physicians' (EPs) manual clinical predictions as a baseline reference.
Reported Outcome: Every machine learning model significantly outperformed human emergency physicians in accurately predicting critical care admissions, general ward admissions, or ED deaths. A fine-tuned Random Forest model demonstrated the most robust predictive performance, and further data analysis explicitly proved that incorporating unstructured text notes via NLP substantially boosted the algorithm's accuracy over using structured data alone.
Stated Limitation: The study relied entirely on historical data collected between 2018 and 2019 across two specific hospital facilities. Although the model underwent successful external validation, it required precise post-modeling threshold adjustments to balance clinical sensitivity against positive predictive value before it could be deemed safely applicable for real-life triage environments.

Okada, Y., Ning, Y., & Ong, M. E. H. (2023). Explainable artificial intelligence in emergency medicine: an overview. Clinical and experimental emergency medicine, 10(4), 354–362. https://doi.org/10.15441/ceem.23.145

Clinical/Engineering Problem: Clinicians naturally lack deep artificial intelligence expertise, leading them to perceive modern machine learning triage tools as a "black box." This creates severe trust and adoption issues in high-stakes emergency environments.
Method Used: A comprehensive literature review mapping out the core definitions, requirements, and categories of Explainable AI (XAI) specifically within emergency medicine. The framework breaks explainability down into three structural tiers: pre-modeling explainability, inherently interpretable models, and post-modeling explainability techniques (such as text justification, simplification, and feature relevance mapping).
Reported Outcome: The review demonstrates that incorporating XAI into emergency care fulfills four critical clinical needs: justification of machine decisions, system control, performance improvement, and medical discovery. Using specific post-modeling tools like text justification allows complex algorithms to communicate clearly with clinical staff.
Stated Limitation: Transitioning these models into live clinical settings faces steep implementation challenges. The paper highlights that current success is bottlenecked by a historical lack of close, cross-disciplinary collaboration between frontline clinicians, system developers, and medical researchers.

Proposed Solution:

This project proposes a four-component digital triage decision support system designed for real-time deployment within the Mercer General Hospital Emergency Department. Each component directly addresses a specific gap identified in the literature, transitioning from the static, single-input models that dominate current research towards a transparent, continuously updating framework built around clinical workflows staff already use.

Component 1: Multi-Source Ingestion Engine

The system ingests patient data through two simultaneous streams rather than a single data entry point. The first stream pulls structured numerical vitals directly from the triage desk monitor, capturing heart rate, blood pressure, respiratory rate, temperature, GCS, and oxygen saturation in real time. The second stream runs a lightweight natural language processing scanner across free-text nurse notes, automatically flagging high-risk clinical language such as "crushing chest pain" or "diaphoretic" to contribute to the risk score even when objective vitals appear borderline. [5] validated this dual-input architecture across 213,984 ED visits at two institutions, demonstrating that every machine learning model tested significantly outperformed human emergency physicians in predicting critical care admissions, and that incorporating unstructured nurse note text via NLP substantially boosted accuracy over structured data alone. [11] corroborated this finding across 20 peer-reviewed papers, confirming that combining free-text and structured data consistently outperforms either stream in isolation.

Component 2: Time-Delta Trajectory Tracking

To eliminate the static snapshot limitation identified by [10] and [8], the system incorporates an automated re-triage clock. If a patient has been waiting longer than 30 minutes without a vital sign update, the dashboard issues a visual prompt to clinical staff. When updated vitals are entered, the system calculates the directional change from arrival values rather than evaluating the new reading in isolation. A respiratory rate that has climbed from 16 to 24 breaths per minute over one hour triggers an immediate priority upgrade, flagging trajectory rather than threshold alone. [8] demonstrated that this temporal approach outperformed static NEWS2 in predicting urgent deterioration within four hours with an AUROC of 0.877 compared to 0.829, catching more critical events at the same false alarm rate. [2] further validated the feasibility of dynamic scoring at scale, developing a continuously updated interpretable scoring protocol across 1.8 million ED visits that evaluated patients at arrival, during their stay, and at admission.

Component 3: Glass-Box Logic Layer

To address the interpretability barrier comprehensively identified by [6] and [3], the system never presents a risk output without an accompanying plain-language clinical explanation. Rather than displaying a raw probability score such as "Risk: 85%", the interface displays the classification mapped directly onto a familiar framework: "High Risk (NEWS2 Score: 6) driven by Respiratory Rate of 25 and Pulse of 110." Every explanation uses clinical terminology staff already recognise rather than algorithmic language they do not. [3] established that XAI increases clinician trust only when explanations are concise, clinically relevant, and directly actionable, and that poorly designed explanations can actively reduce trust or cause clinicians to accept flawed outputs. [9] identified that post-modelling explainability techniques, particularly text justification and feature relevance mapping, are the most effective mechanisms for communicating complex algorithmic reasoning to clinicians in high-stakes emergency environments. [4] validated SHAP-based feature weighting as the appropriate underlying mechanism, achieving a 0.366 increase in average precision over static NEWS2 while maintaining full model auditability across age and sex demographics.

Component 4: Parallel Shadow Workflow Pilot

Over the 12-week pilot window, the system operates entirely in shadow mode alongside Mercer General's existing manual triage processes. Nurses conduct triage exactly as they currently do. The digital system runs silently in parallel, generating its own NEWS2 and SIRS-aligned risk classifications without displaying them to clinical staff or influencing patient care decisions. A research coordinator logs instances where the system's real-time assessment diverged from the manual triage decision, recording the time differential between the system's escalation flag and the point at which clinical staff identified the same deterioration manually. This produces the primary outcome measure for the pilot: the mean time advantage of the automated system over manual re-triage in identifying patients who deteriorated in the waiting room. Taylor et al. (2025) demonstrated in a randomised controlled trial that variability in nurse agreement with AI recommendations makes shadow-mode deployment the safest implementation approach, positioning the technology as a complement to clinical judgement rather than a replacement. This design choice also directly addresses the cross-disciplinary collaboration gap identified by [9] ensuring frontline clinicians, system developers, and researchers are working from the same workflow data before any live deployment decision is made.

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